A MODULE ADMITTING A LEFT DUAL IS LOCALLY A DIRECT SUMMAND OF A FINITE FREE MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The notion of a left dual of an object of a monoidal category, witnessed by a unit and a counit whose composites with the coherence maps are identities. Invoked in the statement (source lines 3089-3091) and in the counted proof (source lines 3105-3116) as a cross-chapter pointer. Supplies the two identities that the proof unfolds at the start.
- The description of the unit as a sum of finitely many products of local sections near a point, and the adjunction between morphisms of modules. Used inside the counted proof at source lines 3117-3128. Produces the finite free module through which the identity factors locally.

Source reference. Original reference categories-definition-dual: sources/stacks-modules/pinned-original-source.tex, source lines 3077–3077. Navigation pointer only: the counted text refers to the Stacks item categories-definition-dual, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference categories-lemma-left-dual: sources/stacks-modules/pinned-original-source.tex, source lines 3145–3145. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-left-dual, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original section example-dual: sources/stacks-modules/pinned-original-source.tex, source lines 3039–3039. Navigation pointer only: the counted text refers to the Stacks item example-dual, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Let $\mathcal{G}, \eta, \epsilon$ be a left dual of $\mathcal{F}$ in the monoidal category of $\mathcal{O}_X$-modules, see Categories, Definition categories-definition-dual. Then*

1. *$\mathcal{F}$ is locally a direct summand of a finite free $\mathcal{O}_X$-module,*
2. *the map $e : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X) \to \mathcal{G}$ sending a local section λ to $(\lambda \otimes 1)(\eta)$ is an isomorphism,*
3. *we have $\epsilon(f, g) = e^{-1}(g)(f)$ for local sections f and g of $\mathcal{F}$ and $\mathcal{G}$.*

Proof. The assumptions mean that

$$\mathcal{F} \xrightarrow{\eta\otimes 1} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{F} \xrightarrow{1\otimes\epsilon} \mathcal{F} \quad \text{and} \quad \mathcal{G} \xrightarrow{1\otimes\eta} \mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \xrightarrow{\epsilon\otimes 1} \mathcal{G}$$

are the identity map. Let $x \in X$. We can find an open neighbourhood U of x, a finite number of sections $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ of $\mathcal{F}$ and $\mathcal{G}$ over U such that $\eta(1) = \sum f_i g_i$. Denote

$$\mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U$$

the map sending the ith basis vector to f_i. Then we can factor the map $\eta|_U$ over a map $\tilde{\eta} : \mathcal{O}_U \to \mathcal{O}_U^{\oplus n} \otimes_{\mathcal{O}_U} \mathcal{G}|_U$. We obtain a commutative diagram

$$\begin{array}{ccccc} \mathcal{F}|_U & \xrightarrow{\eta\otimes 1} & \mathcal{F}|_U \otimes \mathcal{G}|_U \otimes \mathcal{F}|_U & \xrightarrow{1\otimes\epsilon} & \mathcal{F}|_U \\ & \searrow^{\tilde{\eta}\otimes 1} & \uparrow & & \uparrow \\ & & \mathcal{O}_U^{\oplus n} \otimes \mathcal{G}|_U \otimes \mathcal{F}|_U & \xrightarrow{1\otimes\epsilon} & \mathcal{O}_U^{\oplus n} \end{array}$$

This shows that the identity on $\mathcal{F}$ locally on X factors through a finite free module. This proves (1). Part (2) follows from Categories, Lemma categories-lemma-left-dual and its proof. Part (3) follows from the first equality of the proof. You can also deduce (2) and (3) from the uniqueness of left duals (Categories, Remark categories-remark-left-dual-adjoint) and the construction of the left dual in Example example-dual. □